Master

Reflect on Perspective

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0.1 Tools

- ChatGPT: used for grammar editing and improving writing style
- Overleaf: used for writing and formatting the report

0.2 Objects

0.2.1 What kinds of objects, actors, and roles are relevant in your project?

In the context of traffic congestion, several objects and actors are central. Vehicles are at the core and can be divided into human-driven cars and autonomous vehicles. Human-driven cars are operated by drivers with varying habits and reaction times, leading to unpredictable behavior.
This project uses the *Tesla Model Y* as the representative self-driving car, motivated by its position as Tesla's best-selling vehicle and its industry-leading Autopilot system. Central to this is *Tesla Vision*, a camera-only ADAS that uses eight external cameras and neural network processing to monitor the environment, detect objects, and support lane-keeping and speed control[3]. By relying solely on vision-based processing, Tesla Vision reduces driver workload while enhancing safety.
Human drivers influence traffic through decisions on acceleration, braking, and spacing. Road infrastructure provides physical constraints, such as lane numbers, intersections, and overall capacity. Communication mechanisms, including vehicle-to-vehicleand fleet-based data sharing, further enhance Tesla's capabilities. Together, these actors and objects form the system under study. Together, these actors and objects form the system under study.

0.2.2 Which relationships between the objects are relevant

Key interactions include car-to-car distance, which determines whether congestion develops. Close spacing can amplify small disturbances into stop-and-go waves, particularly in human-driven traffic. Tesla-to-Tesla coordination through Autopilot and data sharing smooths braking and acceleration, maintaining optimal spacing and reducing instability. Interactions between Teslas and human-driven cars are also crucial, as mixed traffic conditions limit the extent to which congestion can be prevented, highlighting the transitional nature of traffic systems toward automation.

0.3 Properties

0.3.1 Which properties of the objects do you need to consider

The properties of objects determine how the traffic evolves. For vehicles, speed, braking behavior, and length affect congestion propagation and road capacity. Human drivers introduce a variability of reaction time and driving style, which amplify traffic fluctuations.
For the Tesla Model Y, relevant properties include Tesla Vision's detection accuracy, camera coverage, neural network response, and communication capabilities. Road infrastructure properties, such as lane number, merging points, and speed limits, set baseline conditions for traffic flow. These factors collectively define the system's behavior and provide measurable variables to study how traffic jams form, persist, or may be mitigated by autonomous vehicles.

0.4 Assumptions

0.4.1 Which objects, actors, roles, and properties do you ignore in your project? Give reasons.

In Chapter 2 of Understanding Modelling and Programming, the authors explain that a model is always a simplification of reality. To make a system understandable and reproducible, we must decide which aspects to include and which to ignore[2].
In modelling, a perspective = the viewpoint we choose on reality. It decides which parts of the system matter for our problem. Everything outside that perspective can be ignored [2]. The perspective of our project is to model traffic flow only through the interaction between cars, their spacing, reaction times, and communication.

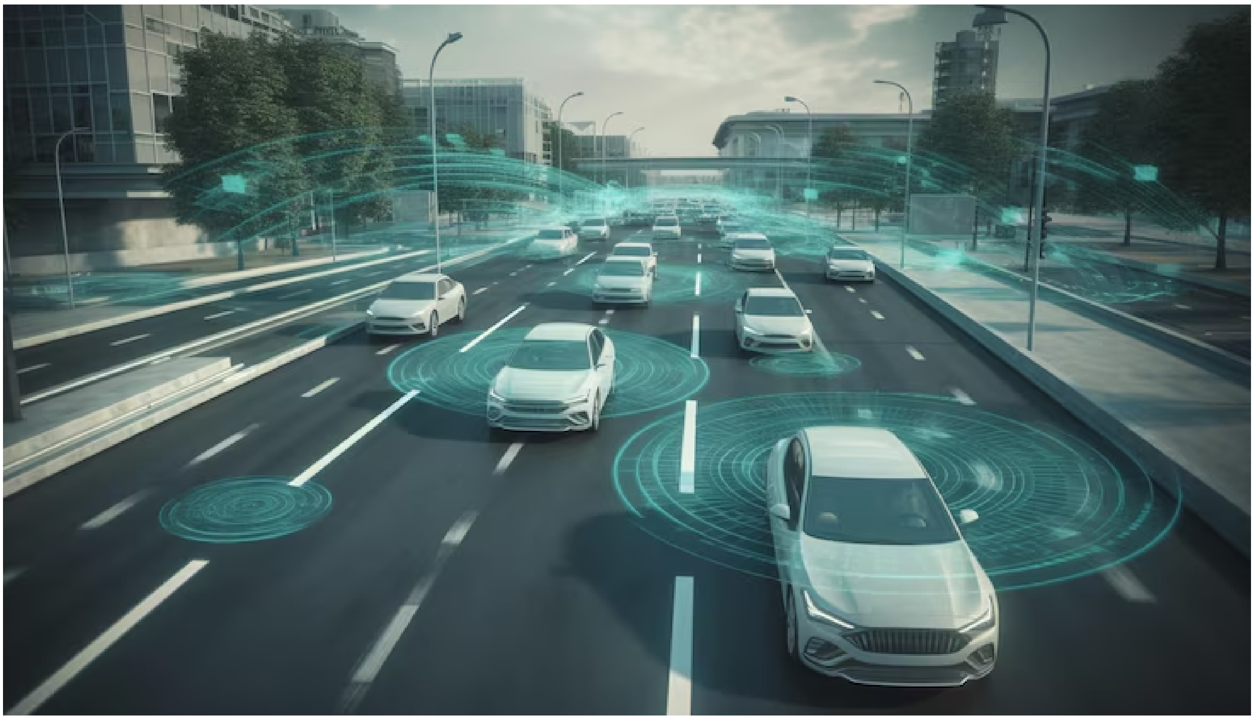

Figure 1: Self Driving Cars
[1]

We ignore weather conditions (rain, snow, fog) to keep the model simple and avoid unnecessary complexity. By removing such external factors, we establish the same baseline situation for all cases, which allows us to better evaluate the difference between human-driven and self-driving cars.
Another factor we exclude is collisions between cars. While collisions can certainly cause traffic jams in reality, modeling them would distract from our main purpose: studying how reaction time and distance between cars influence traffic flow.
We also disregard roadside objects such as trees and infrastructure, since they are static and do not directly contribute to congestion—except in rare cases of extreme weather when a tree might fall into the road. Similarly, humans or animals on the road (e.g., dogs, cats, or birds) are ignored, as these events are unpredictable and outside the scope of our study. Finally, we assume that all vehicles are mechanically sound, meaning no breakdowns occur, so that the model remains focused on driving dynamics rather than technical failures.

In addition, we exclude traffic lights. Although they can be significant sources of congestion in real systems, our model centers on car–car interactions—distance, reaction times, and communication. This simplification narrows the scope, making the model more transparent, reproducible, and aligned with our research question.

This approach follows the principle described in Understanding Modelling and Programming [2]: we include only the aspects of a system that are relevant to the research question, while ignoring details that do not affect the analysis. Just as the book's heating system model considers only radiators, windows, and temperature while leaving out decorations on the wall, our traffic model focuses on vehicles, reaction time, and spacing while ignoring weather, collisions, and other external factors. This ensures that the model remains simple, precise, and reproducible by others.

0.5 Situations

A situation, or a snapshot, is an observation of the system's state at a specific point in time, detailing the configuration of its objects and their attribute values [2]. For our project on traffic jams, a snapshot captures the properties of all *Tesla Model Y* vehicles and the road segment at a single moment.

0.5.1 Situation 1: Free-Flowing Traffic

This is the ideal state of the system, where traffic moves smoothly and efficiently without any impedance.

	Vehicle Attributes
Speed	Vehicles are traveling at or near the maximum legal speed limit.
Distance	There is a significant and safe distance between each vehicle.
Behavior	Acceleration and braking are minimal, with vehicles maintaining a constant velocity.
Communication	V2V communication confirms clear conditions ahead, and data sharing is nominal.
	Road Attributes
Vehicle Density	The number of vehicles on the road is low, far below its maximum capacity.
State	The road is in a state of under-saturation.

Table 1: Attributes for Free-Flowing Traffic.

0.5.2 Situation 2: Transient Congestion

This snapshot captures the critical transition phase where the system moves from free-flowing to congested. This is the point where a "traffic wave" may begin to form from minor events, and where the communication between self-driving cars can be crucial in preventing a full Congestion.

	Vehicle Attributes
Speed	The average speed has dropped significantly below the speed limit and is highly variable between vehicles.
Distance	The distance between vehicles has decreased, leading to tighter spacing.
Behavior	Vehicles exhibit frequent braking and acceleration from reacting to the car directly ahead.
Communication	V2V communication is critical. Vehicles are actively sharing data about reduced speeds and sudden braking, allowing the system to coordinate and dampen the ripple effects to avoid a complete jam.
	Road Attributes
Vehicle Density	The density of vehicles is high, approaching the road's capacity.
State	The road is close to or at the point of saturation.

Table 2: Attributes for Transient Congestion.

0.5.3 Situation 3: Complete Traffic Jam

This situation describes a fully developed traffic jam, where the flow of traffic has come to a near or complete halt.

	Vehicle Attributes
Speed	Vehicle speed is at or near zero.
Distance	Vehicles are stopped with minimal distance between them.
Behavior	Vehicles are idle with occasional short bursts of movement.
Communication	The communication network is used to share status updates and potentially predict the duration of the jam or calculate large-scale rerouting for vehicles that have not yet entered the congested area.
	Road Attributes
Vehicle Density	The density of vehicles has surpassed the road's capacity, reaching maximum or "jam" density.
State	The road is in a state of over-saturation, where traffic throughput is at or near zero.

Table 3: Attributes for a Complete Traffic Jam.

0.6 Describe at least three scenarios (traces) that are relevant to your project.

0.6.1 Scenario 1: Phantom Jam Formation vs. Connected Cars

Description: This scenario illustrates how a small disturbance can escalate into a backward-moving traffic jam without any external cause, such as an accident or a merging point. In a line of human-driven cars, all vehicles initially travel at a steady high speed, around 100 km/h, maintaining what seem to be safe distances. When one driver briefly taps the brakes, perhaps due to distraction or slight overreaction, the effect ripples backwards. The following driver, reacting with a delay, brakes a little harder to remain safe, and each subsequent driver repeats this behavior with increasing intensity. After a few iterations, one car further back is forced to stop completely, creating a “stop wave” that propagates backward and develops into a phantom jam.
In contrast, connected self-driving cars behave differently in the same situation. When a single car detects the need to reduce its speed, it immediately communicates this information to the vehicles behind it. Instead of reacting in isolation and with delay, all the following cars adjust their speeds smoothly and simultaneously. This coordinated response prevents overreactions and absorbs the disturbance before it can grow into a traffic jam.
Relevance: This scenario highlights the core research question of the project: whether vehicle-to-vehicle communication can prevent phantom jams, the most common form of congestion in human-driven traffic.

0.6.2 Scenario 2: Variation in Driver Behavior

Description: This scenario models how differences in human driving styles can create instability in traffic flow. In reality, drivers are not identical: some accelerate aggressively, others more cautiously, and many react with delays of varying length. On a busy highway, these variations accumulate. For example, when one driver accelerates quickly after a slowdown while the car behind reacts more slowly, gaps and sudden braking occur. These inconsistencies propagate backward, generating stop-and-go waves that evolve into traffic jams, even when road conditions are otherwise ideal.
In contrast, connected self-driving cars behave in a uniform and coordinated manner. Reaction time is nearly zero, and acceleration and braking are smooth and synchronized across the platoon. Because the vehicles share information instantly, they avoid the uneven speed adjustments that human variability introduces. As a result, traffic flow remains stable and resistant to the stop-and-go waves that typically form among human-driven vehicles.
Relevance: This scenario illustrates how human variability itself can be a source of congestion. By standardizing behavior and eliminating inconsistent reactions, connected self-driving cars can significantly reduce instability and maintain steady traffic flow.

0.6.3 Scenario 3: Merging at a Bottleneck

Description: This scenario models traffic flow at a fixed physical bottleneck, for example a highway on-ramp where vehicles must merge into the main stream of traffic. In human-driven traffic, this is a frequent point of congestion. The process is often inefficient due to a lack of coordination and drivers on the highway may not adjust their speed to create a gap, or merging drivers may misjudge timing, forcing cars on the main road to brake suddenly. This single braking event disrupts the flow and initiates the stop-and-go waves that characterize congestion. The system transitions from a "free-flowing" state to one of "transient congestion" or a "complete traffic jam" simply because of the inability of independent human agents to efficiently negotiate a shared, limited space. In contrast, connected self-driving cars can manage this bottleneck proactively. As a Tesla on the on-ramp approaches the merge point, it communicates its position, speed, and intent to the platoon of vehicles on the highway. In response, the highway vehicles cooperatively and smoothly adjust their speeds and spacing to form a perfect gap for the merging car to slot into. This "digital zipper merge" occurs without any hard braking or significant speed changes. The system's ability to share data and coordinate action allows it to absorb the merging vehicle while maintaining a stable, free-flowing state, effectively neutralizing the bottleneck.

Relevance: This scenario is relevant because it addresses bottlenecks, the single largest cause of traffic congestion. Unlike the previous scenarios that focused on instabilities arising from driver behavior, this one demonstrates how connected vehicles can manage a permanent, physical constraint on the road infrastructure.

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